CompsNotes: Systems Thinking

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Systems Thinking

1.Fundamental Systems Concepts and Perspectives

Meadows and Wright, 2008 **[pp.1-203]**

Senge, 2006 **[pp. 163-187; 343-349]**

Mental Models

- Mental models: Assumptions, images, stories, "priors"
- Problems when implicit (below conscious awareness)
- If made explcit, create learning organization (tools for personal awareness and reflective skills, institutional practices promote mental models, culture of inquiry and challenge w/o defensiveness or advocacy to win)
- careful with leaps from direct observation to generalization (induction)
- lack of questions = no inquiry
- vulnerability = balance of advocacy and inquiry

Lendaris, 1986

- A Problem Solver must first see a system; no such thing as an objective perspective of a system; An observer must be present to assert systemness; systemness is the focus of the observer; Perception always subject to illusion
- Bias = perceptual filters; a fxn of experience, emotional state, expectations, assumptions, desires, attitudes, social role, etc
- Appropriate bias: perceptions generated by explicit personal reflection and selection of perceptual stance = multiple perspectives: Supra-system (context) + System (focus) + sub-system (parts)
- System: 1) a unit with certain attributes perceived relative to its (external) environment; 2) a unit that has the quality that it internally contains subunits that operate together to manifest the perceived attributes of the unit as a whole.
- "Whole greater than parts": greater than used to bridge two different perceptual levels; attributes at Level A cannot be deduced from attributes at Level B; the whole (Level A) has attributes intrinsic to the joint operation of the parts (Level B) which can only be perceived by going up a level (ie greater than); parts operate by an organizing principle and thus can achieve something greater than the sum of the parts

Bertalanffy, 1984 **[pp. 3-88])**

- The Society for General Systems Research (1954): 1) investigate the isomorphy of concepts, laws and models in various fields; 2) promote the unity of science through improved communication among specialists, help transfer useful ideas between fields; 3) minimize the duplication of efforts
- General Systems research agenda: overcome the limitations of reductionist analytical procedures that assume 1) entity can be resolved into weakly interactive atomics (parts); 2) relations are linear; logico-mathematical foundations for organized complexities (like prob theory is the logico-mathematical foundation for unorganized complexities)
- Approaches: classical (calculus-based) open/closed systems; set theory; graph theory; network theory; cybernetics; information theory; game theory; decision theory; queuing theory; theory of hierarchic order; theory of structure (the order of parts) and function (the order of processes)
- System = organized complexity; nontrivial interactions; a set of simultaneous, coupled differential equations; sets of elements standing in relation; behavior of an isolated element is different than the element embedded in the system
$$\begin{pmatrix} y_1^{(n)} \\ y_2^{(n)} \\ \vdots \\ y_m^{(n)} \end{pmatrix} = \begin{pmatrix} f_1\left(x, \mathbf{y}, \mathbf{y}', \mathbf{y}'', \cdots \mathbf{y}^{(n-1)}\right) \\ f_2\left(x, \mathbf{y}, \mathbf{y}', \mathbf{y}'', \cdots \mathbf{y}^{(n-1)}\right) \\ \vdots \\ f_m\left(x, \mathbf{y}, \mathbf{y}', \mathbf{y}'', \cdots \mathbf{y}^{(n-1)}\right) \end{pmatrix}$$
- Principle of equifinality: open systems allow for a final state to be reached from multiple initial states and different pathways
- Principle of progressive mechanization: initially systems governed by dynamic relationships between components; later fixed arrangements and conditions provide constraints to make the relationships more efficient; this diminishes the equipotentiality of the whole because parts are fixed
- Principle of progressive mechanization plus centralization $\equiv$ biological development; entity moves from a state of interaction between equipotential parts to a state of specialization and subordination under dominant parts
- Advocate for ethics and scientific generalists
- Assume the system (the set of coupled differential equations) is developed into a Taylor series then

1) If the coefficients for all cross terms are zero ⇒ change in each element depends only that element itself ≡ independence of parts; physical summativity; a "heap" complexity; probabilistic accounting applies; only linear terms implies trivial case
2) If the coefficients for all cross terms are not constant but decrease with time ⇒ the whole split into independent causal chains; progressive segregation; progressive mechanization; results from increased specialization driven by energy flux of open systems; loss of regulation and potentiality and resilience (parts become irreplaceable if damaged)
3) If the coefficients for one or two terms increase relative to all others ⇒ progressive centralization; a centralized system that may experience massive change from small inputs to the central variables; leads to organisms ontogenetically (organism development) and to spp phylogenetically (spp development)
4) If elements become increasingly specialized ⇒ hierarchical order; each element becomes a system in itself

- Systems that attain a final state (stationary as Le Chatelier) appear to have teleological character where the system seems to "aim" at the final state. In fact, just a distance from equilibrium

- The exponential law and the logistic law are cited as isomorphisms that support a general systems theory b/c they apply across fields

zwick_elements_2014 [pp. 60-71])

- System: a nexus of structure / function / history; elements and relations (constraints from randomness)

- Model: a set of elements having attributes linked by relations (a system)

- Mario Bunge's epistemological hierarchy:
  1) observables
  2) relations, laws, hypotheses
  3) models
  4) theories

- Holistic analysis of a system requires consideration of structure and function and history. Structure and function are synchronic (at the same time); History is diachronic (at different times)

- Be aware of simplistic diachronic models (linear change, cyclicity, random drift) and more complex models (nonlinear dynamics, non-equilibrium thermodynamics, generalized evolution) where initial conditions or internal fluctuations may be random but subsequent unfolding is lawful

- What is structural and what is functional depends on the scale at which a phenomenon is viewed which depends on the focal point (the vertex) of the system

- Systems range from the extreme where structure is far more important than function (materials science) to the extreme where function is far more important than structure (

- Structural analysis (reductionist) is often privileged over function

- Complexity results from:
  1) Higher ordinality relations (greater than diadic or triadic relations)
  2) Non-linear relations
  3) Mutual branching of causality (multiple causes per effect; feedback)
  4) Global causality or the sensitivity and vulnerability of a system (highly networked)
  5) Necessity (determinism) and chance (stochasticity) are both present
  6) Perceptual stance: Course-grained (less complex b/c of aggregation) vs. fine-grained (all elements and relations accounted for)
  7) Hierarchical coupling: If levels of a hierarchy are well-insulated then each level is a whole unto itself. Otherwise, coupling naturally creates multi-scalar entities

- Emergence is the opposite of reduction ⇒ Reduction looks down from L to L1; Emergence looks up from L1 to L

Ashby, 1955 **[pp. 86-117, 195-224])**
Simon, 1996 **[pp. 169-216])**
Barabási and Bonabeau, 2003
Barabási, 2009
Crutchfield et al., 1986
mitchell_complexity_2009 [pp. 94-111; 169-185]
Nicolis and Prigogine, 1989 **[pp. 5-44; 183-192; 217-242]**
Cover and Thomas, 2006 **[pp. 1-30; 34-35; 57-60]**

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